

Whole Exome Sequencing Analysis

Patient ID: XX	PIN: XX
Age: 28 Years	Gender: Male
Hospital/Clinic: XX	Sample Number: XX
Referring Clinician: XX	Sample Collection Date: NA
Specimen: DNA	Sample Received Date: NA
	Report Date: NA

Clinical History

Proband, XX is born of a non-consanguineous marriage. XX is presented with young-onset diabetes mellitus. On examination, BMI is 17.1 kg/m². C-peptide levels are low, GAD and IA-2 are negative. XX is currently being treated with insulin. Proband, XX has been evaluated for pathogenic variations.

Results

Variant of uncertain significance related to the given phenotype was identified

List of uncertain significant variant(s) identified:

Gene	Variant*	Allele Status	Region	Disease or Disorder	Inheritance	Classification*
<i>HNF1B</i> (-)	c.689A>T (p.Lys230Met)	Heterozygous	Exon 3	Renal cysts and diabetes syndrome (OMIM# 137920)	Autosomal Dominant	Uncertain Significance (PM2_Supp, PP3)

*Genetic test results are based on the recommendation of American college of Medical Genetics [1].

No other variant that warrants to be reported for the given clinical indication was identified

Variant Interpretation:

Gene	Variant*	Inheritance	Disease or Disorder	Criteria	Classification*
<i>HNF1B</i> (-)	c.689A>T (p.Lys230Met)	Autosomal Dominant	Renal cysts and diabetes syndrome (OMIM# 137920)	PM2_Supp, PP3	Uncertain Significance
Region	Allele Status	Sequence Ontology	Variant Frequency	Allelic Read Depths	
Exon 3	Heterozygous	Missense	Not observed in annotated gnomAD v4 All	Ref(T): 100, Alt(A): 92, VAF: 47.91%	
In silico Prediction Tools				Conservation Status	
SIFT: Da	LRT: D	PolyPhen2: Da	CADD: D	MT2: D	PhyloP: C GERP++: C
Genomic Position			Clinical & Literature Evidences		
Chr17:NC_000017.11:g.37733677T>A (NM_000458.4)			-		

Abbreviation: **D:** Deleterious; **Da:** Damaging; **Ds:** Disrupted; **C:** Conserved; **T:** Tolerated; **U:** Uncertain; **Supp:** Supporting

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OMIM Phenotype: Renal cysts and diabetes syndrome (OMIM#[137920](#)) is caused by heterozygous mutation in the *HNF1B* gene (OMIM#[189907](#)). Renal cysts and diabetes syndrome (RCAD) is an autosomal dominant multisystemic disorder with significant phenotypic heterogeneity. It is characterized by (1) nondiabetic renal disease resulting from abnormal renal development, and (2) diabetes, which in some cases occurs earlier than age 25 years and is thus consistent with a diagnosis of maturity-onset diabetes of the young (MODY). The renal disease is highly variable and includes renal cysts, glomerular tufts, aberrant nephrogenesis, primitive tubules, irregular collecting systems, oligomeganephronia, enlarged renal pelvises, abnormal calyces, small kidney, single kidney, and horseshoe kidney. This disease follows autosomal dominant pattern of inheritance [2].

Additional Variant(s):

The additional variants identified which are significant but may not be related to patient's phenotype.

Phenotype – genotype correlation is recommended.

List of additional uncertain significant variant(s) identified:

Gene	Variant*	Allele Status	Region	Disease or Disorder	Inheritance	Classification*
<i>LRP6</i> (-)	c.4429A>G (p.Lys1477Glu)	Heterozygous	Exon 21	{Coronary artery disease, autosomal dominant, 2} (OMIM# 610947)	Autosomal Dominant	Uncertain Significance

Variant Interpretation:

Gene	Variant*	Inheritance	Disease or Disorder	Criteria	Classification*
LRP6 (-)	c.4429A>G (p.Lys1477Glu)	Autosomal Dominant	{Coronary artery disease, autosomal dominant, 2} (OMIM#610947)	PM2_Supp, PP3	Uncertain Significance
Region	Allele Status	Sequence Ontology	Variant Frequency	Allelic Read Depths	
Exon 21	Heterozygous	Missense	Not observed in annotated gnomAD v4 All	Ref(T): 85, Alt(C): 98, VAF: 53.55%	
In silico Prediction Tools				Conservation Status	
SIFT: Da	LRT: D	PolyPhen2: Da	CADD: U	MT2: D	PhyloP: C GERP++: C
Genomic Position			Clinical & Literature Evidences		
Chr12:NC_000012.12:g.12125316T>C (NM_002336.3)			-		

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No other variant that warrants to be reported for the given clinical indication was identified.

CNV Findings

No significant Copy Number Variations (CNV) related to phenotype was detected.

Recommendations

- Sequencing the variant(s) in the parents and the other affected and unaffected members of the family is recommended to confirm the significance.
- Alternative test is strongly recommended to rule out the deletion/duplication.
- Genetic counselling is advised.

Methodology

DNA extracted from the blood was used to perform whole exome using whole exome capture kit. The targeted libraries were sequenced to a targeted depth of 80X to 100X using SURFSeq 5000 sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels. Sentieon DNAScope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (v4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. In-silico predictions of the variant was carried out using VS- SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported. Gene list used in the assay [Click here](#).

Sequence data attributes

Total Read Generated	16.14 Gb
Data ≥ Q30	94.14%

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed.
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non-disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

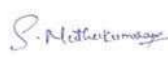
Disclaimer

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR and Promoter region variants are not assessed using this assay.
- This assay has a sensitivity of 70-75% in detecting large deletions/duplications of more than 10 base pairs or copy number variations (CNV). However, it is important to note that any CNVs detected must be confirmed using an alternate method.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.
- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions. The variants of uncertain significance and variations with high minor allele frequencies which are likely to be benign will be given upon request.
- In addition, this test does not cover detection of such structural variants in the genes *SMN1*, *SMN2*, *HBA1*, *HBA2*, *PMS2*, *GBA*, and *CYP21A2* which have very high homology. Moreover, the ORF15 transcript isoform (RefSeq id: NM_001034853.1) of the *RPGR* gene associated with retinitis pigmentosa is not sufficiently covered in the test.
- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [3].

References

1. Richards, S, et al. Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genetics in medicine: official journal of the American College of Medical Genetics. 17.5 (2015): 405-424.
2. Amberger J, Bocchini CA, Scott AF, Hamosh A. McKusick's Online Mendelian Inheritance in Man (OMIM). Nucleic Acids Res. 2009 Jan;37(Database issue):D793-6. doi: 10.1093/nar/gkn665. Epub 2008 Oct 8.
3. Lee, Kristy et al. "ACMG SF v3.3 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG)." Genetics in medicine: official journal of the American College of Medical Genetics vol. 27,8 (2025): 101454. doi:10.1016/j.gim.2025.101454.

This report has been reviewed and approved by:

				
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Appendix 1: Gene Coverage

Indication Based Analysis:

Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered	Gene Name	Percentage of coding region covered
ABCA1	100.0	ABCC8	100.0	ABCG5	100.0	ABCG8	100.0
ACAT1	100.0	ACSF3	100.0	ADCY3	100.0	AFF4	96.97
AGL	100.0	AGPAT2	100.0	AGRP	100.0	AIP	100.0
AIRE	100.0	AKT2	100.0	ALDOB	100.0	ALG1	100.0
ALG11	100.0	ALG12	100.0	ALG3	100.0	ALG6	100.0
ALG8	100.0	ALG9	100.0	ALMS1	100.0	APOA1	100.0
APOA5	100.0	APOB	100.0	APOC2	100.0	APOC3	100.0
APOE	100.0	APPL1	100.0	AQP2	100.0	ARL13B	100.0
ARL6	100.0	ARMC5	100.0	ATP6V0A2	100.0	AVP	100.0
AVPR2	100.0	B4GALT1	100.0	BBS1	100.0	BBS10	100.0
BBS12	100.0	BBS2	100.0	BBS4	100.0	BBS5	100.0
BBS7	100.0	BBS9	100.0	BDNF	100.0	BLK	100.0
BSCL2	100.0	CANT1	100.0	CAV1	100.0	CAVIN1	100.0
CC2D2A	100.0	CCDC28B	100.0	CEL	94.67	CEP164	100.0
CEP19	100.0	CEP290	100.0	CETP	100.0	CHD2	100.0
CIDEC	100.0	CISD2	100.0	CNOT1	100.0	COG1	100.0
COG4	100.0	COG5	100.0	COG6	100.0	COG7	100.0
COG8	100.0	CP	100.0	CPE	100.0	CREBBP	100.0
CUL4B	100.0	CYP27A1	100.0	DCAF17	100.0	DNAJC3	100.0
DOLK	100.0	DPM1	100.0	DPM2	100.0	DPM3	100.0
DUT	100.0	DYRK1B	100.0	EHMT1	100.0	EIF2AK3	100.0
EIF2B1	100.0	EIF2S3	100.0	ENO3	100.0	ENPP1	100.0
EPM2A	100.0	FBN1	100.0	FBP1	100.0	FOXP3	100.0
GAA	100.0	GATA4	100.0	GATA6	100.0	GBE1	100.0
GCK	100.0	GCKR	100.0	GH1	100.0	GHR	100.0
GHRHR	100.0	GHRL	100.0	GLI3	100.0	GLIS3	100.0
GLUD1	100.0	GMPPA	100.0	GNAS	100.0	GNE	100.0
GPC3	100.0	GPD1	100.0	GPIHBP1	100.0	GYS1	100.0
GYS2	100.0	H6PD	100.0	HADH	100.0	HEXA	100.0
HMGCL	100.0	HMGCS2	100.0	HNF1A	100.0	HNF1B	100.0
HNF4A	100.0	HSD11B1	100.0	IER3IP1	100.0	IFT172	100.0
IFT27	100.0	IGF1R	100.0	IL2RA	100.0	INPP5E	100.0
INSR	100.0	ITCH	100.0	KCNJ11	100.0	KCNJ6	100.0
KIDINS220	100.0	KIF7	100.0	KLF11	100.0	KMT2C	100.0
LAMP2	100.0	LARGE1	100.0	LAS1L	100.0	LDHA	100.0
LDLR	100.0	LDLRAP1	100.0	LEP	100.0	LEPR	100.0
LIPA	100.0	LIPC	100.0	LIPE	99.77	LMF1	100.0
LMNA	100.0	LPL	100.0	LRBA	100.0	LZTFL1	100.0
MAFA	100.0	MAGEL2	100.0	MAN1B1	100.0	MANF	100.0
MC3R	100.0	MC4R	100.0	MEGF8	100.0	MFN2	100.0
MGAT2	100.0	MKKS	100.0	MKRN3	100.0	MKS1	100.0
MNX1	96.19	MOGS	100.0	MPDU1	100.0	MPI	100.0
MPV17	100.0	MRAP2	100.0	MTNR1B	100.0	MTTP	100.0
MTX2	100.0	MYO5A	100.0	MYO7A	100.0	MYT1L	100.0
NDN	100.0	NEUROD1	100.0	NEUROG3	100.0	NGLY1	100.0
NHLRC1	100.0	NKX2-2	100.0	NPHP1	100.0	NPHP3	100.0
NROB2	100.0	NSD1	100.0	NTRK2	100.0	OFD1	100.0
OTULIN	100.0	OXCT1	100.0	PAX4	100.0	PAX6	100.0
PC	100.0	PCBD1	100.0	PCK1	100.0	PCNT	100.0
PCSK1	100.0	PCSK9	100.0	PDE11A	100.0	PDE4D	100.0
PDX1	100.0	PFKM	100.0	PGK1	100.0	PGM1	100.0
PHF6	100.0	PHIP	100.0	PHKA1	100.0	PHKA2	100.0
PHKB	100.0	PHKG2	100.0	PIK3R1	100.0	PLAGL1	100.0
PLIN1	100.0	PMM2	100.0	PNPLA6	100.0	POLD1	100.0
POMC	100.0	PPARG	100.0	PPP1R15B	100.0	PRKAG2	100.0

PRKAR1A	100.0	PRKCE	100.0	PRMT7	100.0	PROM1	100.0
PRPH2	100.0	PSMB8	100.0	PTEN	100.0	PTF1A	100.0
PYGL	100.0	PYGM	100.0	RAB23	100.0	RAI1	100.0
RBCK1	100.0	RDH5	100.0	RFT1	100.0	RFX6	100.0
RHO	100.0	RLBP1	100.0	RPGRIP1L	100.0	RPS6KA3	100.0
SDCCAG8	100.0	SETD2	100.0	SH2B1	100.0	SIM1	100.0
SLC16A1	100.0	SLC19A2	100.0	SLC29A3	100.0	SLC2A2	100.0
SLC35A1	100.0	SLC35A2	100.0	SLC35C1	100.0	SNRPN	100.0
SPG11	100.0	SRD5A3	100.0	SSR4	100.0	STAT1	100.0
STAT3	100.0	TBX3	100.0	THOC2	100.0	THRA	100.0
TMEM165	100.0	TMEM67	100.0	TRAF3IP1	100.0	TRAPP9	100.0
TRMT10A	100.0	TTC21B	100.0	TTC8	100.0	TUSC3	100.0
UCP2	100.0	UCP3	100.0	VPS13B	100.0	WFS1	100.0
WRN	100.0	XRCC4	100.0	XYLT1	100.0	ZBTB20	100.0
ZFP57	100.0	ZMPSTE24	100.0	ZNF711	100.0	LRP6	100.0